

Name:

Section:

Criterion D Summative Assessment: Applying Mathematics in Real-Life Contexts

Project Title: “*The Scale of Allah’s Creation*”

“Indeed, We have created all things with precise measure.”

— Surah Al-Qamar 54:49

إِنَّا كُلَّ شَيْءٍ خَلَقْنَاهُ بِقَدَرٍ
— سورة القمر ٥٤:٤٩

Transliteration: Innā kulla shay’in khalaqnāhu biqadar.

This verse beautifully expresses Allah’s precision and perfect order in creation.

Task Overview

Allah created the universe with balance, order, and precision. In this project, you will explore the scale of His creation using **scientific notation** and **mathematical comparisons**. You’ll reflect on how mathematics helps us recognize Allah’s wisdom and perfect design.

Part 1: Selecting and Recording Measurements

Level 1 and 2 Questions:

Choose three real-world measurements from Allah's creation.

Examples: The diameter of an atom, diameter of human cell, the Earth's diameter, the distance to the Moon, the diameter of an average size naval orange, the diameter of an average size of an apple, the diameter of a watermelon, the distance between the earth and the sun, the distance between the sun and the moon, ext.

Directions: Ensure you have the correct Units for both forms for all 3 creations.

Object of Allah's Creation	Measurement (Standard Form)	Measurement (Scientific Notation)	Unit
1.(A) Antartica	4,000 km	4×10^3	KM
2.(B) Atom	0.0000000001 km	1×10^{-10}	KM
3.(C) UY Scuti	4,735,000,000 km	4.735×10^6	KM

Level 3 and 4 AQuestions:

Part 2: Comparing Measurements Using Scientific Notation

Use **multiplication** or **division** to compare the scales of your measurements.
Show all your steps clearly.

Example: How many times larger is the diameter of Earth than a human cell?

$$\frac{1.27 \times 10^7}{7 \times 10^{-6}} = 1.8 \times 10^{12}$$

The Earth's diameter is 1.8×10^{12} times larger than a human cell.

Comparison	Work Shown	Answer in Scientific Notation
A vs. B You can change the ratio from A:B to B:A according to which number is bigger than the other.	$4 \times 10^3 \div 1 \times 10^{-10} = 4 \times 10^3 - (-10)$	4×10^{13}
B vs. C You can change the ratio from B:C to C:B according to which number is bigger than the other.	$4.735 \times 10^9 / 1 \times 10^{-10} = 4.735 \times 10^9 - (-10)$	4.735×10^{19}
A vs. C You can change the ratio from A:C to C:A according to which number is bigger than the other.	$4.735 \times 10^9 / 4 \times 10^3 = 1.18375 \times 10^6$	1.18375×10^6

Extra calculations:

What is the sum of the diameter of Earth and the diameter of the Moon?
Write the final answer in scientific notation form. Show your work

$$1.27 \times 10^4 + 3.48 \times 10^3 = 16.18 \times 10^3 = 1.618 \times 10^4 \text{ km}$$

What is the difference between the diameter of Earth and the diameter of the Moon? Show your answer in scientific notation form.

$$1.27 \times 10^4 - 3.48 \times 10^3$$

$$12.7 \times 10^3 - 3.48 \times 10^3 = 9.22 \times 10^3$$

$$\text{Ans } 9.22 \times 10^3$$

Level 5 and 6 Questions: Be Creative

Part 3: Visual Representation

Create a **chart, diagram, or digital poster** that shows your three measurements **to scale**.

You may draw by hand. If you want colored construction paper, let me know and I will be more than happy to provide you with it. Be creative!

My Visual Representation:

On a separate paper. Make sure to be proportional with your drawing of Allah Creations to match with your findings. For example, if you find the diameter of the Earth to the diameter of the Sun, the diameter of the sun is greater by double, then draw the sun double the size of the earth.

Level 7 and 8 Questions:

Part 4: Reflection

Write a short paragraph explaining how your findings show Allah’s precision and wisdom in creation.

You may begin with:

“From studying these results, I recognized Allah’s precision and wisdom in creation because...”

Reflection:

“From studying these results, I recognized Allah’s precision and wisdom in creation because even the smallest things, like atoms, and the largest things, like suns and Antarctica, have specific measurements and sizes. Comparing these measurements helped me see how organized and detailed Allah’s creation is. It reminds me that everything in the universe has a purpose and is created with wisdom.”

Qur’anic or Hadith Connection:

The Qur’an says:

رَدَقَبْ هُنَّ قُلُوبُ عِيَالِكِ أَنْ

Inna kulla shay'in khalaqnahu bi-qadar

“Indeed, We created everything with precise measure.”

— Qur’an 54:49 Surah Al-Qamar

Criterion D Rubric: Applying Mathematics in Real-Life Context.

Level	Description
1 and 2	Identifies some relevant measurements but may have errors in data or units. Applies basic scientific notation with limited accuracy. Communicates work with minimal clarity.
3 and 4	Identifies relevant measurements and applies scientific notation with partial accuracy. Uses some appropriate methods for multiplying or dividing in scientific notation. Communicates work that is adequately organized, though with minor errors.
5 and 6	Selects accurate and relevant measurements, applies scientific notation correctly, and uses appropriate operations to solve the problem. Communicates solutions clearly in an adequately organized format. Reflection identifies a link between precision in measurements and divine creation.
7 and 8	Selects precise and relevant measurements, applies scientific notation accurately and efficiently, and uses appropriate operations without error. Communicates solutions clearly, thoroughly, and in a well-organized manner. Reflection thoughtfully discusses how the precision and order in these measurements demonstrate the wisdom and design in Allah's creation.

Criterion C: Communications

Questions Levels	Students is able to do
1 & 2	i. use limited mathematical language ii. use limited forms of mathematical representation to present information iii. communicate through lines of reasoning that are difficult to interpret.
3 & 4	i. use some appropriate mathematical language ii. use appropriate forms of mathematical representation to present information adequately iii. communicate through lines of reasoning that are able to be understood, although these are not always clear iv. adequately organize information using a logical structure.
5 & 6	i. usually use appropriate mathematical language ii. usually use appropriate forms of mathematical representation to present information correctly iii. move between different forms of mathematical representation with some success iv. communicate through lines of reasoning that are clear although not always coherent or complete v. present work that is usually organized using a logical structure.
7 & 8	i. consistently use appropriate mathematical language ii. use appropriate forms of mathematical representation to consistently present information correctly iii. move effectively between different forms of mathematical representation iv. communicate through lines of reasoning that are complete and coherent v. present work that is consistently organized using a logical structure.